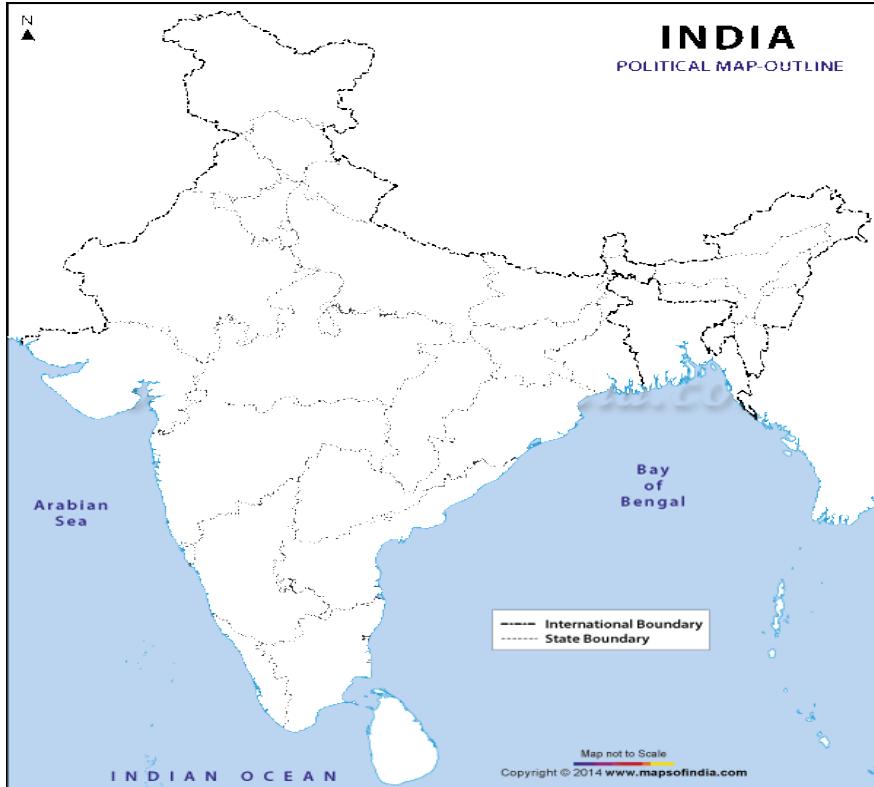




Sovereign Cloud

Unified Control Management Platform

Data sovereignty will become increasingly critical for Government Departments to mitigate geopolitical risks



Data sovereignty is enforced through DPDP, IT Act, CERT-In rules, and government cloud frameworks that together ensure Indian data stays under Indian jurisdiction and control

Sovereign Control Principle

For sensitive government and security data:
Data must remain in India
Infrastructure must be under Indian control

Data sovereignty means that all sensitive national and citizen data remains physically within India, fully controlled by Indian authorities, and governed solely by Indian laws — with no foreign jurisdiction, control plane, or dependency involved

What is sovereign cloud and why do we need it?

What is Sovereign Cloud?

The ability of nations, governments, or organizations to develop, deploy, and control artificial intelligence (AI) technologies.

All data resides and is controlled within the jurisdiction of the sovereign nation.

Operations and management of cloud infrastructure are carried out by **citizens of that sovereign nation**.

Foreign authorities cannot assert authority over the data stored in the sovereign cloud.

Why do we need Sovereign Cloud?

Control Over Technology:

Reduce nations dependency foreign cloud providers for critical technology infrastructure.

Public Cloud Lock-In:

~42% are concerned about critical data being managed by public cloud providers

Increasing Data Volumes:

95% of businesses cite need to manage unstructured data locally as a problem, which sovereign cloud can address.

Increasing Regulations:

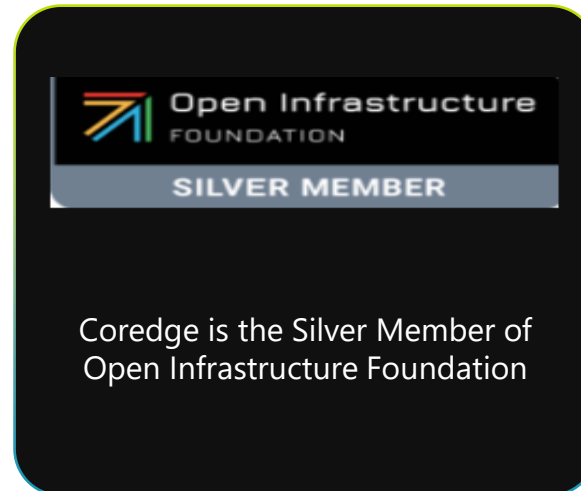
The ever-changing landscape of legal requirements and regional laws necessitates sovereign control over data.

Reduced Operational Cost:

By controlling the energy consumption, which is a significant contributor to cloud pricing, sovereign cloud can provide affordable cloud services.

A Testament to Our Commitment to Quality and Innovation

Our industry recognitions and certifications are a testament to our ability to consistently deliver innovative solutions that meet the highest standards of quality and performance.



Coredge Platforms



Coredge Cloud Suite (CCS)

Comprehensive Cloud Management Suite including any virtual machine management(Rhel,Ubuntu,KVM),any container management(Docker,AKS,GKE,,), storage orchestration, multi-cloud management, and advanced orchestration capabilities.



Cirrus Cloud Platform (CCP - CVM+CKP+Cloud Orchestrator)

Comprehensive platform including virtual machine management, storage orchestration, multi-cloud management, and advanced orchestration capabilities



Cloud Orbiter (CO+Ckp)

Gateway to manage multi-cluster Kubernetes orchestration to manage clusters, containers, and edge systems, ensuring seamless interoperability and multi-cloud orchestration..



Dflare AI

Comprehensive AI Cloud Offering to provide GPU-as-a-service, Inferencing-as-a-service and AI workspaces-as-a-service. Build complete AI offering using Dflare.

Core Components of CCS

CCS is a comprehensive solution comprising several core components that work in tandem to deliver full-stack cloud management:



Unified Dashboard

Single UI and API for all cloud resource control.



Identity & Access Management (IAM)

Built-in SSO and RBAC for secure access.



Compute & Storage Orchestration

Automates VM/container provisioning and scaling.



Networking & Load Balancing

Manages virtual networks and ensures app performance.



Database as a Service (DBaaS)

Automates database deployment and scaling.



Monitoring & Observability

Real-time metrics and analytics for uptime.



Orchestration & Automation Engine

Drives workflows, auto-scaling, and self-healing.

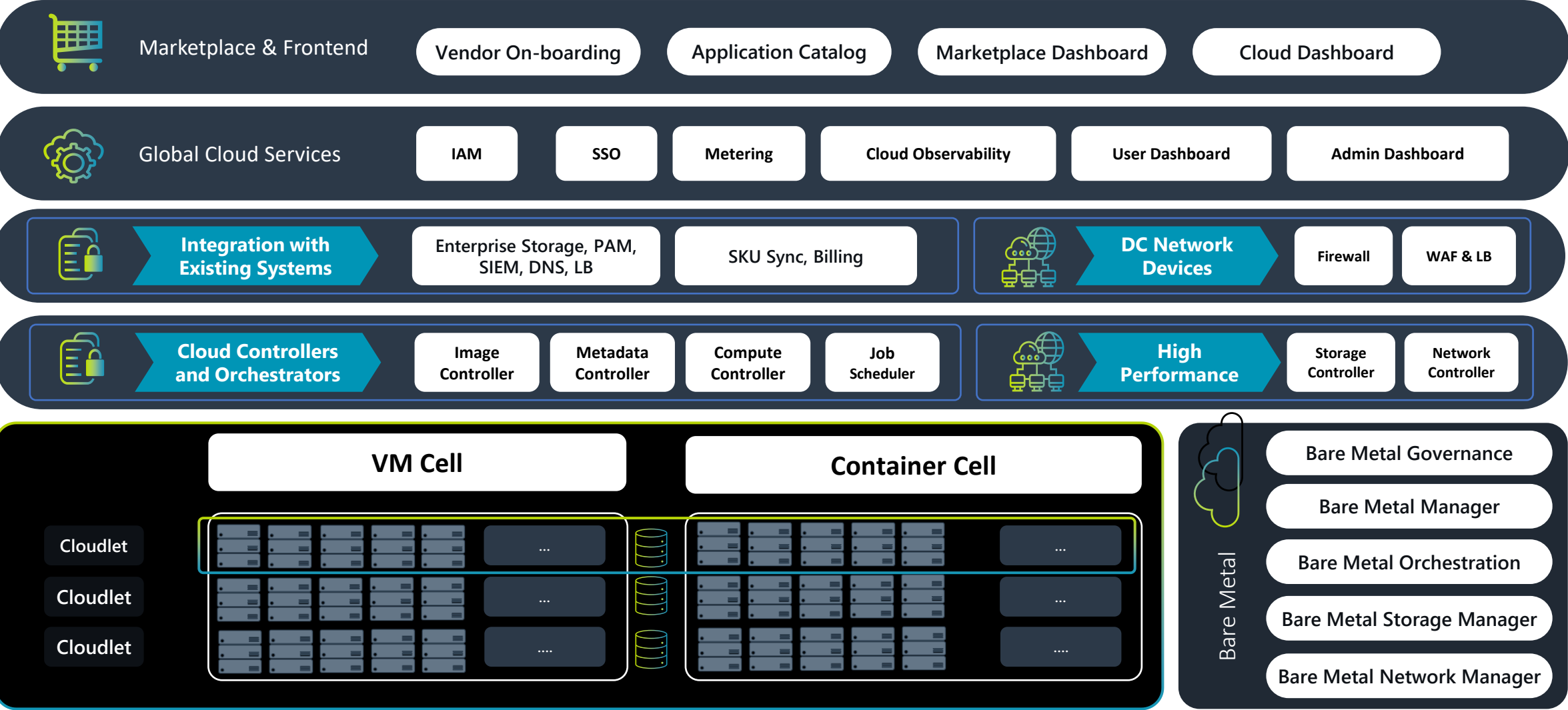


Multi-Cloud Integration Layer

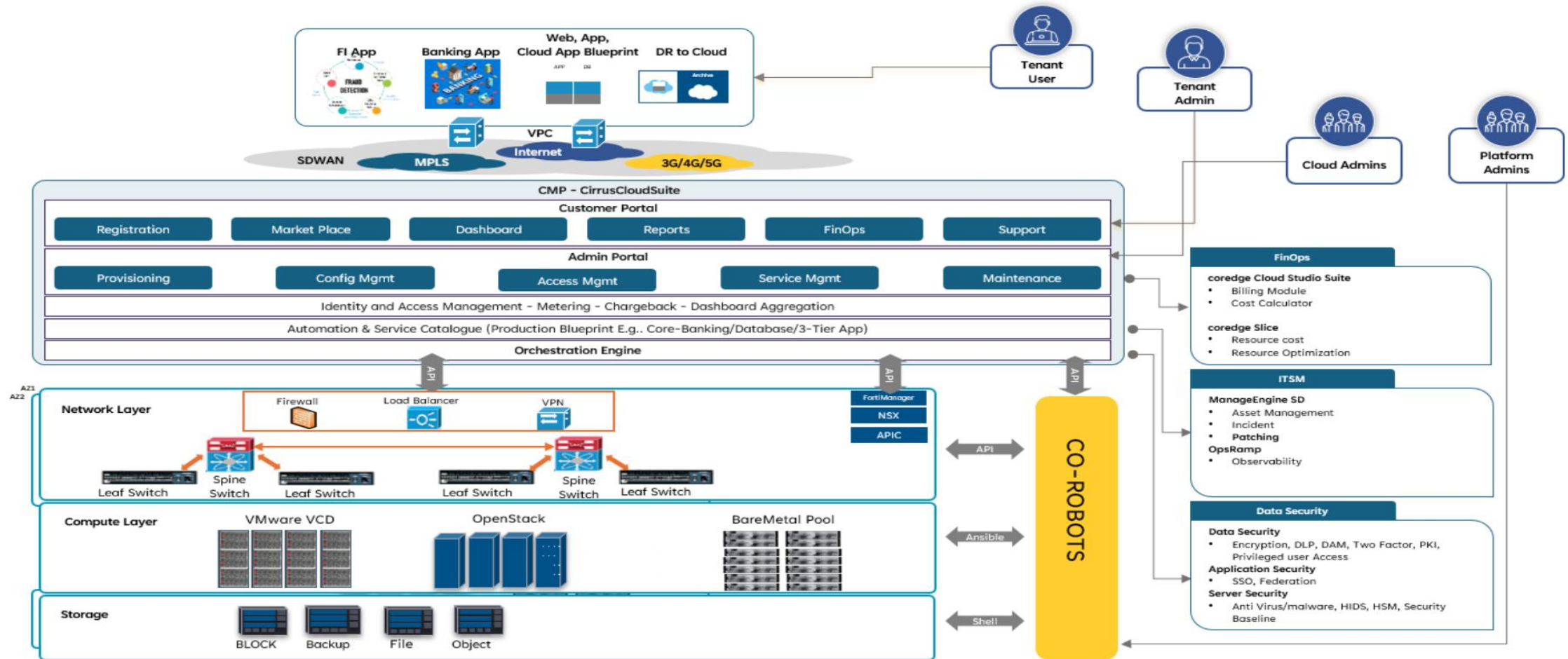
Natively connects AWS, Azure, and OpenStack.

Bottom Line: CCS replaces multiple point tools with one unified platform for cloud management, monitoring, orchestration, security, and cost control.

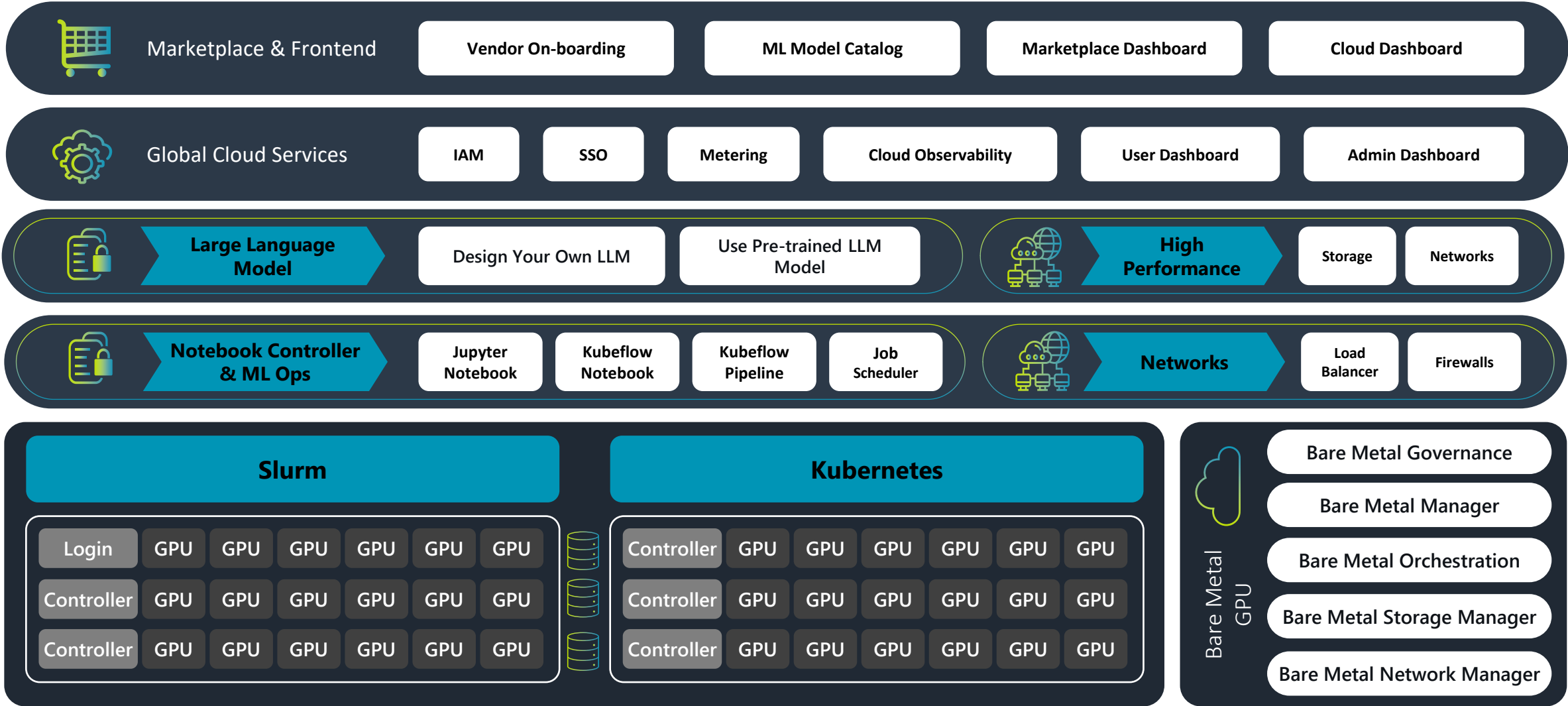
Coredge Sovereign Cloud



CCS Integration Architecture



AI Cloud As-a-service building blocks



CSP Admin Portal

Region : DC-1 Panchkula

Search



Last Login:
22:09:15 | 12/08/2024

HOME

 Dashboard

 Provider Dashboard

ADMINISTRATOR

 Tenants

 User Management

 Quota Manage... >

 Monitoring >

 Access Logs

CATALOG

 Service Catalog

PROVIDERS

 Cloudlets

 Storage Providers

 Backup Providers

 ITSM Provider

 CMDB Providers

 Baremetal Providers

 Provider Clusters

Welcome to CSP Cloud Admin Portal

Designed for seamless management and complete control, the Admin Portal empowers you to oversee users, roles, and resources effortlessly, with secure access and real-time insights that simplify administration and enhance operational efficiency.



Get started with CSP Cloud Admin

To get started, you'll need to connect your first cloud provider. This will enable you to manage virtual machines, monitor resources, track usage, and control your entire cloud infrastructure from this centralized dashboard.

Ready to Get Started?

Connect your cloud provider in just a few simple steps



Cloudlets

A cloud provider that delivers scalable, on-demand infrastructure and services for your applications.

 Quick setup process

 Secure API authentication

Connect Now

+ Add Cloudlet

****A Cloudlet is a single, independent OpenStack cloud environment that is onboarded into the CMP as one logical unit**

- **One OpenStack cluster = One Cloudlet**
- **One Cloudlet = One Region inside CMP**

CSP Admin Portal – Service Catalogue

CSP

Region : DC-1 Panchkula

Search

Last Login: 22:09:15 | 12/08/2024

HOME

Dashboard

Provider Dashboard

ADMINISTRATOR

Tenants

User Management

Quota Manage...>

Monitoring>

Access Logs

CATALOG

Service Catalog

PROVIDERS

Cloudlets

Storage Providers

Backup Providers

ITSM Provider

CMDB Providers

Baremetal Providers

Provider Clusters

Dashboard>Service Catalog>Service Catalog

Service Catalog

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+ Add New Service

Our Services

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Baremetal

Dedicated physical server with no virtualization layer—ideal for high-performance, latency-sensitive, and compute-heavy workloads. Offers full control and consistent performance.

Container Registry

A secure, scalable storage for managing and sharing container images. Easily push, pull, and version images for seamless CI/CD and deployment.

Kubernetes

A managed container orchestration platform for deploying, scaling, and managing containerized applications. Automates rollouts, scaling, and resource optimization.

Load Balancers

Distributes incoming traffic across multiple servers to ensure high availability, scalability, and optimized performance for your applications.

Public IP

A static IP address that allows external access to your cloud resources over the internet. Essential for hosting publicly reachable services.

Virtual Machines

A scalable compute instance with virtualized resources for running applications, workloads, or operating systems in the cloud.

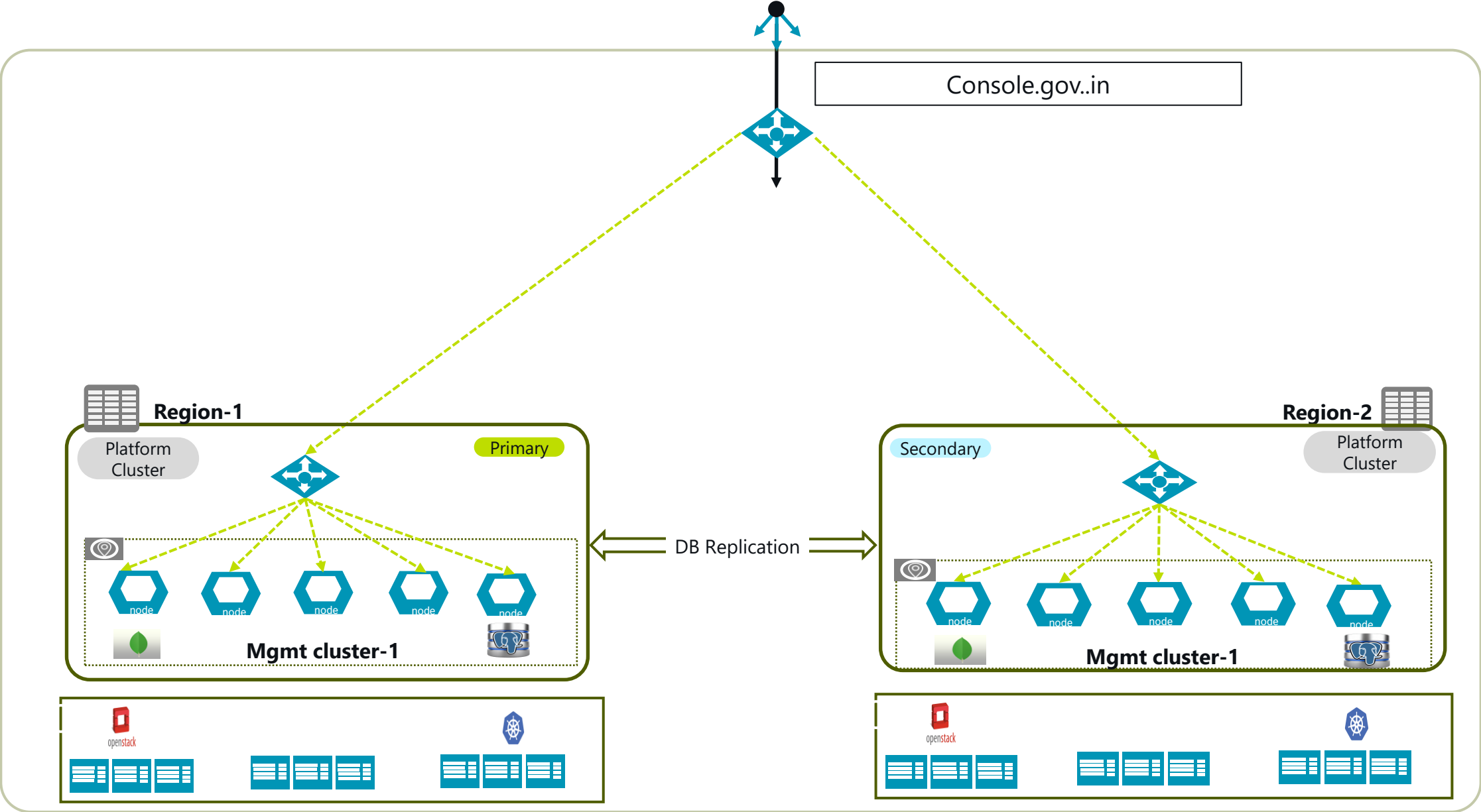
Volumes

Storage resources used to retain data across compute instances. Suitable for

VPN

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CMP Orchestration Architecture



Powered by Coredge DFlare(GPU Orchestrator) & DELL 9680

Cloud and AI Infra



AI Platforms(API & PASS)

- **Background :** Govt outside is building a nation-scale AI ecosystem. To power this, they partnered with DELL Technologies + Coredge to create a high-performance, sovereign AI cloud built for massive GPU operations.
- **Challenges :**
 - Enforcing data sovereignty & compliance
 - Operating 12,000+ GPUs for AI/ML training
 - Accelerating time-to-market for AI infrastructure services
- **Solution : DELL Technologies — AI Infrastructure Backbone**
 - PowerEdge XE9680:** Ultra-dense, high-performance AI server
 - Engineered for extreme GPU workloads + high-bandwidth fabrics
 - Built-in security & silicon trust → sovereign-by-design platform
 - Proven reliability for hyperscale AI clusters
- **Outcome :** A jointly engineered **DELL + Coredge Sovereign AI Cloud** delivering:
 - Secure, compliant national AI infrastructure
 - Massive GPU scale with intelligent orchestration
 - Faster service launch with lower operational burden



**12,000
+ GPUs**

**DELL XE
9680**

**DELL
Powerscale**

Cloud Services provided by Coredge platform

Compute	Storage	Containers	Managed Database	Platform as a Service	Hybrid and Multi-Cloud
- Baremetal - Composable compute - General purpose compute - High availability compute - Storage optimized compute - CPU for VMs	- Block Storage - File storage - Object storage - Data fabric - Archival services	- Managed Kubernetes - Observability for Kubernetes - Elastic Container Registry (Private Registry) - Helm and Operator support Application Deployment	- PostgreSQL - Redis - MySQL/MariaDB	- HPC - Lab and Environment as a Service - Developer Runtime Environment (Java, Python, Ruby on Rails etc) - Devops as a Service	- Basic GitOps - Observability for Cloud Native - Monitoring as a Service - Messaging Services - Notification Service
Edge	Migration	Networking	Security	Risk & Compliance	Data Protection
- Edge Device Manager - Edge Orchestration - Ruggedized Deployment	- Application Migration - Data Migration - Lift & Shift Migration	- Software Defined Networking - Load Balancer as a Service - IX - CDN - SD-WAN - IP-Transit	- Unified IAM - Certificate Manager and Secrets - Virtual Private Environment - Firewall and Web Application Firewall - Key Management	- Continuous compliance - Cost visibility - Software license and asset management - Cluster Certification	- Backup & Recovery - Automated DR - Managed backup - Snapshot backup for App & DB

Our Customers



Customer	Applications	Scale (CPU/vCPU/GPU)	Time Period
 Govt of India (NIC) Sovereign Cloud	MyGov Websites, S3WaaS, DBIMToolkit, Gov.in portal	20,000+ vCPUs, 2,200+ VMs, 10 OpenStack clusters, 7 CEPH clusters	In Production
 Govt Enterprise Cloud (Power Discom)	Consumer Meter Data Mgmt, Smart Meter Cloud, Real-time Billing	100 + nodes	In Production
 Tier-1 Telco (India) Internal Cloud (CellStack)	Self-service provisioning, Secure PAM, Real-time Monitoring, Flexible IaaS & PaaS services	8,000+ Cores	In Production
Tier-1 Telco (India) B2B Enterprise Cloud	IaaS (VMs, Storage, LB, Security), PaaS (Kubernetes, Container Registry, DBaaS)	500+ nodes	In Production
 Kenya National Healthcare Cloud	National Healthcare Registry, Insurance Claims, Fraud Analytics, ERP for Claims	12,000 vCPUs	Stable for 1+ year
 Abu Dhabi IHC Cloud	Sovereign Blockchain, Internal IT workloads, Oracle DB Clusters, AI/ML workloads	8,000 vCPUs + 1000 +GPUs	In Production

Thanks